A Constant-One Finite-Determination Subcase

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Status. Partial result, likely valid. This packet proves a subcase of the sharp finite-determination problem from arXiv:2307.15143; it does not settle the general problem.

Source Problem

The source paper is F. Catrina and M. I. Ostrovskii, *Finite determination for embeddings into Banach spaces: new proof, low distortion*, arXiv:2307.15143. The relevant source question is Problem 1 in Section 4, page 15:

corresponding types of embeddability are also finitely determined.

4 Related open problems

In our opinion, the most important open problem related to the topic of this paper is:

Problem 1. *Can one replace $(3+\epsilon)$ by $(1+\epsilon)$ in Theorem 1.6?*

This problem has been already raised in [OO19, Problem 5.1]. Both positive or negative answer would be very interesting.

Another interesting problem is mentioned in Remark 1.4. For that problem it is not clear whether it is realistic to describe completely the corresponding class of Banach spaces.

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In words, the problem asks whether the $(3+\varepsilon)$ distortion in Theorem 1.6 can be replaced by $(1+\varepsilon)$. The source notes that this was already raised as Problem 5.1 in Ostrovska–Ostrovskii.

Definitions

Definition 1. *A Banach space X is* separably isometrically universal *if every separable Banach space admits a linear isometric embedding into X.*

By the Banach–Mazur theorem, $C[0,1]$ is separably isometrically universal.

Proof Intuition

The standard finite-determination argument already gives an embedding of the locally finite metric space into an ultrapower of X with no loss in the distortion. The loss in general comes from pulling that image back from the ultrapower to X. If X contains every separable Banach space isometrically, there is no loss: a locally finite metric space is countable, so the closed linear span of its ultrapower image is separable and embeds isometrically into X.

Theorem 1. *Let X be separably isometrically universal. Let (M, d) be a locally finite metric space. Suppose that every finite subset of M admits a bi-Lipschitz embedding into X with distortion at most C. Then M admits a bi-Lipschitz embedding into X with distortion at most C.*

Proof. Fix a base point $O \in M$. Since M is locally finite, $M = \bigcup_{n=1}^{\infty} B(O, n)$ is countable. For each finite subset $N \subset M$ containing O, choose a map $f_N : N \to X$ with $f_N(O) = 0$ and, after rescaling,

$$d(u, v) \le \|f_N(u) - f_N(v)\| \le C\, d(u, v) \qquad (u, v \in N).$$

Let $\mathcal{U}$ be an ultrafilter on the directed set of finite subsets of M containing O and containing every tail $\{N : N \supset N_0\}$. Define $f : M \to X^{\mathcal{U}}$ by

$$f(u) = (f_N(u))_{\mathcal{U}},$$

where for coordinates with $u \notin N$ one may put $f_N(u) = 0$; the tail condition makes this choice irrelevant in the ultrapower. For every $u, v \in M$, the set of N containing $\{O, u, v\}$ lies in $\mathcal{U}$, hence

$$d(u, v) \le \|f(u) - f(v)\|_{X^{\mathcal{U}}} \le C\, d(u, v).$$

Let $E = \overline{\operatorname{span}}\, f(M) \subset X^{\mathcal{U}}$. Since M is countable, E is separable. By separable isometric universality there is a linear isometry $J : E \to X$. The map $J \circ f : M \to X$ satisfies the same two-sided estimates, and therefore has distortion at most C. □

Corollary 1. *$C[0, 1]$ has constant-one finite determination: if all finite subsets of a locally finite metric space embed into $C[0, 1]$ with distortion at most C, then the whole metric space embeds into $C[0, 1]$ with distortion at most C.*

Proof. Apply the theorem and the Banach–Mazur theorem. □

Limitations and Novelty Check

This result rules out separably isometrically universal spaces, including $C[0, 1]$, as witnesses to a strict finite-determination gap. It does not prove the $(1 + \varepsilon)$ version of Theorem 1.6 for arbitrary Banach targets, and it does not construct a space X with $D(X) > 1^+$.

The local cheap indexes showed an existing constant-one packet for $L_p[0, 1]$ and no packet for $C[0, 1]$ or the separably universal mechanism. Bounded web/arXiv searches for exact phrases around $D(C[0, 1])$, finite determination, separably universal spaces, and locally finite metric spaces did not reveal an explicit prior answer to this subcase.

References

[1] F. Catrina and M. I. Ostrovskii, *Finite determination for embeddings into Banach spaces: new proof, low distortion*, arXiv:2307.15143.

[2] S. Ostrovska and M. I. Ostrovskii, *Distortion in the finite determination result for embeddings of locally finite metric spaces into Banach spaces*, Glasgow Math. J. 61 (2019), 33–47.